

JEE MAIN REVISION MASTER: AREA UNDER THE CURVES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 17 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Area Under Curves & Between Intersecting Parabolas	17 PYQs	Integral Applications

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Area Under Curves & Between Intersecting Parabolas	$\text{Area between } y^2=4ax \text{ and } x^2=4by = \frac{16ab}{3}$, $\text{Area between } y^2=4ax \text{ and } y=mx = \frac{8a^2}{3m^3}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Area Under Curves & Between Intersecting Parabolas (Very High Yield — 17 PYQs)

Core Principle: Area bounded by curve $y = f(x)$, x -axis and lines $x=a$, $x=b$: $A = \int_a^b y \, dx$. Area enclosed between two parabolas $y^2 = 4ax$ and $x^2 = 4by$ is directly $(16ab / 3)$. Area between parabola $y^2 = 4ax$ and line $y = mx$ is $(8a^2 / 3m^3)$. Area of ellipse $(x^2 / a^2) + (y^2 / b^2) = 1$ is πab .

■■ **Caution / JEE Pitfall:** If the region falls below the x -axis ($y < 0$), the integral gives a negative value; area is always positive, so take the absolute value $|\int y \, dx|$!

■ **Speed Trick:** Memorize the standard KCET shortcuts: (1) Area between $y^2=4ax$ and $x^2=4ay$ is $(16a^2 / 3)$. (2) Area between $y^2=4ax$ and $y=mx$ is $(8a^2 / 3m^3)$. Solves the problem in 5 seconds.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The area of the region bounded by the curve $y^2=x^3$, the y -axis and the lines $y=1$ and $y=8$ is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The area bounded by the curve $y=\sin((x / 3))$, x axis, the lines $x=0$ and $x=3\pi$ is..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The area of the region bounded by the curve $y=x^2$ and the line $y=16$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The area of the region bounded by the curve $y^2 = x^3$, the y -axis and the lines $y=1$ and $y=8$ is

- (A) $(155 / 3)$ sq. units
- (B) $(93 / 5)$ sq. units
- (C) 93 sq. units
- (D) 155 sq. units

Correct Answer: None

Solution Analysis: The curve is $y^2 = x^3$. Since the region is between the lines $y=1$ and $y=8$, it is easier to write x in terms of y . From $y^2 = x^3$ we get $x = y^{2/3}$. The y -axis is $x=0$. So, for each value of y from 1 to 8, the horizontal width of the region is $x_{\text{right}} - x_{\text{left}} = y^{2/3} - 0 = y^{2/3}$. Hence, the required area is $\int_1^8 y^{2/3} dy$. Now, $\int y^{2/3} dy = \frac{y^{5/3}}{5/3} = \frac{3}{5} y^{5/3}$. So, Area = $\left[\frac{3}{5} y^{5/3} \right]_1^8 = \frac{3}{5} (8^{5/3} - 1^{5/3})$. Since $8^{1/3} = 2$ we have $8^{5/3} = 2^5 = 32$. Thus, Area = $\frac{3}{5} (32 - 1)$.